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Perceived Impact of School Plant Management on Academic Staff Job Performance in Public Universities in South-East, Nigeria

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Abstract

This study investigated the perceived impact of school plant management on academic staff job performance in public universities in South-East, Nigeria. The purpose of the study was to determine the perceived impact of school plant management on academic staff job performance in public universities in South-East, Nigeria. Two research questions guided the study and two hypotheses were tested at 0.05 level of significance. Descriptive survey research design was adopted for the study. The population consisted of 12,540 academic staff from 15 public universities in South-East, Nigeria. A sample of 627 academic staff members representing 5% of the population from 15 public universities in South-East, Nigeria was used for the study. Multistage sampling procedure was used to select the sample for study. The instrument used for data collection was a questionnaire titled "School Plant Management and Academic Staff Job Performance Questionnaire (SPMASJPQ)" with a reliability coefficient of 0.90. Mean Scores and Standard Deviation were used for answering the research questions. Chi-square test of goodness-of-fit was used to test the hypotheses at 0.05 level of significance. The findings of the study revealed that school plant management in the areas of classroom facilities and office spaces have significant positive impact on academic staff job performance in public universities in South-East, Nigeria. It was concluded that school plant management positively impacts on academic staff job

performance in public universities in South-East, Nigeria. It was therefore, recommended among others that university administrators should upgrade research facilities by procuring modern laboratory equipment and improving research support units through strategic budgeting. Better research environments will strengthen staff output, innovation and scholarly performance.

Keywords: school plant management, academic staff job performance, research facilities, ICT infrastructure

Introduction

Academic staff job performance is globally recognized as a fundamental driver of university excellence, influencing student learning, research output and institutional reputation. However, challenges such as poor lecture delivery, low research productivity and inadequate student engagement have been observed in various higher education systems. In advanced countries, declining research output in some United states universities and teacher burnout in the United Kingdom seems have affected institutional performance (Özdemir & Gören, 2017). In Nigeria, public universities appear to face similar issues, including ineffective lecture delivery, poor research contributions and administrative inefficiencies. In South-East, Nigeria, concerns persist over overcrowded classrooms, inadequate research facilities and poor ICT infrastructure, all of which seems to hinder academic staff effectiveness. These challenges could be linked to school plant management, as conducive learning and working environments, adequate office spaces, research facilities and stable power supply can enhance academic staff job performance, ultimately fostering academic staff job performance in public universities.

Academic staff job performance refers to the effectiveness with which university lecturers and researchers carry out their teaching, research, administrative and community service responsibilities (Shaikh, Saad, & Bhutto, 2018). Furthermore, it encompasses instructional delivery, research output, student mentorship, participation in academic committees and contribution to institutional development (Adzongo, Yiman & Mbam, 2025). Effective academic staff job performance is crucial for achieving high student outcomes, knowledge advancement and global competitiveness in higher education (Danasabe, 2018). However, challenges such as inadequate resources, poor motivation and unfavorable working conditions often hinder optimal performance, particularly in developing countries like Nigeria (Nnebedum & Akinfolarin, 2017). Studies have shown that a well-structured and conducive work environment significantly influences academic staff productivity (Bua, 2023). This underscores the role of school plant management in shaping the conditions that facilitate effective academic staff job performance.

School plant management involves the planning, maintenance and effective utilization of physical facilities, including classrooms, offices, research laboratories, libraries, ICT centers and other infrastructural elements that support teaching and learning (Bua, 2024). Proper management of school facilities enhances academic activities by providing an enabling environment for instructional delivery, research and administrative duties (Ukpong & Uchendu, 2016). Inadequate school plant management has been linked to inefficiencies in academic staff job performance, such

as poor lesson delivery, reduced research output and low staff motivation (Gable, Lee, Kwahk, & Green, 2022).

Effective school plant management ensures regular maintenance, proper allocation of spaces and adequate provision of essential utilities, such as electricity and internet access (Bonnici & Centigo, 2019). Given that academic staff productivity is largely influenced by the quality of their work environment, improving school plant management in universities can significantly enhance academic staff job performance. Therefore, the study is focused on examining the perceived impacts of research facilities and information and communication technology infrastructure on academic staff job performance in public universities.

Research facilities include laboratories, libraries, funding opportunities and access to academic journals and equipment essential for scholarly work (Cambridge Assessment International Education, 2017). High-quality research facilities enhance knowledge creation, publication output and innovation among academic staff (Nnebedum & Akinfolarin, 2017). In South-East, Nigerian universities, research productivity is hindered by inadequate research laboratories, outdated equipment and limited access to digital resources (Afolabi, Adewale & Ojo, 2020). Weakened school plant management could contribute to the neglect of research facilities, restricting lecturers' ability to conduct meaningful investigations and publish their findings (Bonnici & Centigo, 2019). Poor research infrastructure could limit funding opportunities, collaborations and technological advancements. Ensuring well-equipped and accessible research facilities is crucial for improving academic staff performance (Ezeokoli, Nwankwo & Okafor, 2022). Similarly, information and communication technology (ICT) infrastructure is essential for modern teaching, research and administrative efficiency in universities.

Information and communication technology (ICT) infrastructure encompass computers, internet access, digital learning platforms and software that facilitate communication, research and administrative work in universities (Oni, Nwajiuba, & Nwosu, 2017). Adequate ICT resources enhance academic staff job performance by enabling online teaching, research collaboration and access to global knowledge repositories (Oladipo, Ajayi & Oyetunde, 2021). However, weak school plant management in South-East, Nigerian universities seem to have resulted in inadequate ICT infrastructure, slow internet connectivity, outdated software and insufficient ICT training for staff (Gable, Lee, Kwahk, & Green, 2022). These deficiencies could limit lecturers' ability to integrate technology into teaching and research, reducing efficiency and innovation (Nwankwo, Okeke & Udeh, 2022). Strengthening ICT infrastructure is essential for improving the quality of education and administrative processes as well as ensure a secure campus environment which is critical for academic staff job performance.

In recent years, the condition of school facilities in public universities across South-East, Nigeria has become increasingly alarming. Research laboratories seem to lack basic equipment and frequent power outages appear to disrupt academic activities. Lecturers often struggle to teach, conduct research, or access digital resources in such deteriorating environments. If these issues persist, the academic quality and productivity of universities in the region may further decline, making this study both timely and urgently necessary.

Statement of the Problem

Academic staff job performance appears to determine educational quality and institutional effectiveness worldwide, as it influences student learning, research output and overall university rankings. In Nigeria, concerns have been raised by educational stakeholders, including university administrators, policymakers and students, about the declining performance of academic staff in public universities. Nnebedum and Akinfolarin (2017) point out issues such as reduced research productivity, ineffective teaching delivery, inadequate student engagement and delays in administrative responsibilities to affect academic staff job performance. These concerns could be linked to poor working conditions, lack of motivation and inadequate infrastructure in universities.

In public universities in South-East, Nigeria, the researcher has observed critical issues affecting academic staff job performance, including insufficient research facilities, unstable power supply and limited access to information and communication technology (ICT) infrastructure. These inadequacies, which fall under the domain of school plant management, seems to create an unconducive environment for academic staff to effectively carry out their responsibilities. When essential facilities such as research centers and ICT infrastructures are poorly managed, academic staff experience difficulties in lecture delivery, research and administrative tasks, leading to diminished job performance.

Efforts have been made by relevant stakeholders, including the National Universities Commission (NUC), the Tertiary Education Trust Fund (TETFund) and university governing councils, to improve academic staff job performance through infrastructural interventions, professional development programmes and increased funding (Ministry of Education, 2020). However, despite these measures, persistent issues such as inadequate maintenance of facilities, insufficient workspaces, poor security and unreliable power supply continue to hinder optimal performance among academic staff. These ongoing challenges seems to highlight the inefficiency of school plant management in public universities in South-East, Nigeria. Given the crucial role of school plant management in shaping the work environment of academic staff, this study seeks to investigate the perceived impact of school plant management on academic staff job performance in public universities in South-East, Nigeria.

Purpose of Study

The purpose of the study was to investigate the perceived impact of school plant on academic staff job performance in public universities in South-East, Nigeria. Specifically, the study sought to investigate;

1. the perceived impact of provision of research facilities on academic staff job performance in public universities in South-East, Nigeria.
2. The perceived impact of provision of information and communication technology infrastructure on academic staff job performance in public universities in South-East, Nigeria.

Research Questions

The following research questions guided the study.

1. What is the perceived impact of provision of research facilities on academic staff job performance in public universities in South-East, Nigeria?
2. What is the perceived impact of provision of information and communication technology infrastructure on academic staff job performance in public universities in South-East, Nigeria?

Hypotheses

The following null hypotheses were formulated and tested at 0.05 level of significance.

1. Provision of research facilities have no significant impact on academic staff job performance in public universities in South-East, Nigeria.
2. Provision of information and communication technology infrastructure has no significant impact on academic staff job performance in public universities in South-East, Nigeria.

Methodology

The study adopted the descriptive survey research design. According to Emaikwu (2022), descriptive survey research design, collects detailed descriptions of existing phenomena to justify current conditions and practices or to make better plans for improving existing situations. The design was considered appropriate because the study investigated respondents' perceptions without manipulation of variables. It also enables the researcher to collect data from a larger population and generalize the findings.

The population consisted of 12,540 academic staff from 15 public universities in South-East, Nigeria. A sample of 627 academic staff members representing 5% of the population from 15 public universities in South-East, Nigeria was used for the study. Multistage sampling procedure was used to select the sample for study. The instrument used for data collection was a questionnaire titled "School Plant Management and Academic Staff Job Performance Questionnaire (SPMASJPQ)". The instrument was validated by three experts and it yielded a reliability coefficient of 0.90. The instrument was administered to the respondents by the researcher with the help of seven research assistants. Mean Scores and Standard Deviation were used for answering the research questions. Chi-square test of goodness-of-fit was used to test the hypotheses at 0.05 level of significance.

The Chi-square (χ^2) test of goodness of fit was used for testing hypotheses at 0.05 alpha level of significance. The reason for using Chi-square was to compare the expected result with the actual result to determine whether there is statistically significant difference between observed data and expected data. The decision rule for rejection or otherwise of the hypotheses was based on p value and alpha value. Hypotheses of no significant impact was not rejected for any hypothesis whose p value was equal to or greater than alpha value of 0.05 ($p \geq 0.05$). While any hypothesis whose p value was less than alpha value of 0.05 ($p \leq 0.05$) was rejected.

Results

Question One: What is the perceived impact of provision of research facilities on academic staff job performance in public universities in South-East, Nigeria?

Table 1:

Mean and Standard Deviation Scores of the Perceived Impact of Provision of Research Facilities on Academic Staff Job Performance in Public Universities in South-East, Nigeria.

S/No	Items Description	N	SA	A	D	SD	$\bar{x}$	STD	Decision
1	Well-equipped research laboratory enhances my research productivity.	608	112	304	164	28	2.82	0.78	Agree
2	Research funding opportunities increase my publication output.	608	117	401	71	19	3.01	0.66	Agree
3	Available research infrastructure fosters collaborative research projects.	608	133	385	65	25	3.03	0.70	Agree
4	Timely access to journal updates my course content.	608	97	421	67	23	2.97	0.65	Agree
5	Specialized fieldwork facilities improve my research quality.	608	126	365	94	23	2.98	0.72	Agree
6	Modern data analysis tools enhance my research findings.	608	118	374	98	18	2.97	0.69	Agree
7	Research facilities improves the quality of my publications.	608	126	392	70	20	3.03	0.67	Agree
8	Adequate research facilities make research supervision more effective.	608	112	394	87	15	2.99	0.65	Agree
Cluster Mean and Standard Deviation							2.98	0.69	Agree

Source: Researchers' Field Survey Results (2026)

Table 1 reveals that items 17-24 have Mean values of 2.82, 3.01, 3.03, 2.97, 2.98, 2.97, 3.03 and 2.99 for items 1–8, with standard deviation of 0.78, 0.66, 0.70, 0.65, 0.72, 0.69, 0.67 and 0.65. All the mean scores exceed the 2.50 cut-off mark, showing that respondents agreed with every item relating to research facilities. These findings point to the conclusion that well-equipped laboratories, funding opportunities, research infrastructure, timely access to journals, specialized fieldwork facilities, modern data analysis tools and adequate research facilities all significantly improve staff research productivity, collaborative projects, publication quality and supervision effectiveness. The cluster mean of 2.98 illustrates high overall agreement that research facilities

strongly support academic staff performance, while the cluster standard deviation of 0.69 indicates strong consistency in responses. Consequently, the data clearly demonstrates that the provision of quality research facilities plays a critical role in strengthening academic staff job performance in public universities in South-East, Nigeria.

Question Two: What is the perceived impact of provision of information and communication technology infrastructure on academic staff job performance in public universities in South-East, Nigeria?

Table 2:

Mean and Standard Deviation Scores of the Perceived Impact of Provision of Information and Communication Technology Infrastructure on Academic Staff Job Performance in Public Universities in South-East, Nigeria.

S/N	Items Description	N	SA	A	D	SD	$\bar{x}$	STD	Decision
9	Reliable internet connection supports my access to academic resources.	608	80	324	141	63	2.69	0.83	Agree
10	Updated computer system enhances my lecture preparation.	608	67	393	90	58	2.77	0.77	Agree
11	ICT resources facilitate effective visual classes.	608	114	255	126	113	2.61	0.99	Agree
12	E-resources enable me to update lecture materials regularly.	608	107	255	121	125	2.57	1.01	Agree
13	Online assessment tools improve students' evaluation effectively.	608	100	212	195	101	2.51	0.96	Agree
14	ICT tools support productive virtual meetings.	608	112	398	75	23	2.99	0.68	Agree
15	Technical ICT support accelerate issue resolution.	608	101	405	73	29	2.95	0.69	Agree
16	ICT infrastructure enhances my research output.	608	97	240	226	45	2.64	0.84	Agree
Cluster Mean and Standard Deviation							2.72	0.85	Agree

Source: Researchers' Field Survey Results (2026)

Table 2 reveals that items 9–16 recorded mean scores of 2.69, 2.77, 2.61, 2.57, 2.51, 2.99, 2.95 and 2.64, with standard deviations of 0.83, 0.77, 0.99, 1.01, 0.96, 0.68, 0.69 and 0.84. All items meet the minimum mean threshold of 2.50, showing that respondents agreed that ICT infrastructure positively influences academic staff job performance. The results suggest that reliable internet access, updated computer systems, ICT resources for visual classes, e-resources

for material updates, online assessment tools, ICT support for virtual meetings, technical support systems and ICT-enhanced research output all contribute meaningfully to staff efficiency and teaching quality. The cluster mean of 2.72 confirms the respondents' overall agreement on the positive effect of ICT facilities, while the cluster standard deviation of 0.85 indicates moderate uniformity in their responses. These findings imply that ICT infrastructure is an essential driver of academic staff job performance in public universities in South-East, Nigeria, enabling improved teaching, research and administrative tasks.

Hypothesis 1: Provision of research facilities have no significant impact on academic staff job performance in public universities in South-East, Nigeria.

Table 3:

Chi-square Test of the Impact of Provision of Research Facilities on Academic Staff Job Performance in Public Universities in South-East, Nigeria.

Responses	SA	A	D	SD	Df	χ^2	P-value	Remark
Observed Frequency	118	380	89	21				
					3	488.62	0.000	Sig.
Expected Frequency	152	152	152	152				

$P=.000 < 0.05$; $df=3$

Table 3 shows that $P < .05$ at 3 degree of freedom. Therefore, the null hypothesis which states that provision of research facilities have no significant impact on academic staff job performance in public universities in South-East, Nigeria was rejected. This means that, provision of research facilities have significant positive impact on academic staff job performance in public universities in South-East, Nigeria.

Hypothesis 2: Provision of information and communication technology infrastructure has no significant impact on academic staff job performance in public universities in South-East, Nigeria.

Table 4:

Chi-square Test of the Impact of Provision of Information and Communication Technology Infrastructure on Academic Staff Job Performance in Public Universities in South-East, Nigeria.

Responses	SA	A	D	SD	Df	χ^2	P-value	Remark
Observed Frequency	97	310	131	70				
					3	231.28	0.000	Sig.
Expected Frequency	152	152	152	152				

$$P=.000<0.05; df=3$$

Table 4 shows that $P<.05$ at 3 degree of freedom. Thus, the null hypothesis which states that provision of information and communication technology infrastructure has no significant impact on academic staff job performance in public universities in South-East, Nigeria was therefore, rejected. The implication is that provision of information and communication technology infrastructure has significant positive impact on academic staff job performance in public universities in South-East, Nigeria.

Discussion of Findings

The findings of the study are discussed as follows:

The first finding of the study revealed that provision of research facilities has significant positive impact on academic staff job performance in public universities in South-East, Nigeria. This finding is supported by high mean ratings (2.82–3.03) across items 1–8, where respondents agreed that well-equipped laboratories facilitate research, availability of research funding increases publication output, timely access to journals updates course content and modern data analysis tools improve research interpretation. These responses confirm that research facilities directly influence lecturers' scholarly productivity and teaching quality. This finding aligns with Afolabi, Adewale and Ojo (2020), who reported a significant positive relationship between research facilities such as laboratories, grants and internet access and research output of academic staff in South-West, Nigeria. Their study is relevant because both examine research facility availability as a determinant of academic productivity, though in different regions, demonstrating that the impact of research infrastructure is consistent across Nigeria. Similarly, Ezeokoli, Nwankwo and Okafor (2022) found that research facility provision significantly predicted academic staff engagement, publication output and conference participation in federal universities in South-East, Nigeria, which directly mirrors the present study's emphasis on how research laboratories, access to journals and research supervision tools enhance job performance. Their study is especially relevant because it shares the same regional context and focuses on institutional support for research productivity. The justification for this finding lies in the nature of academic work itself: meaningful research output depends on access to up-to-date facilities, reliable data sources, laboratory equipment and research funding. Without such tools, lecturers face barriers to publishing, supervising students and contributing to global knowledge, making research facilities indispensable for academic performance. Within the scope of the study, it is found that provision of research facilities support greatly the academic staff job performance in public universities in South-East, Nigeria.

The second finding of the study revealed that provision of information and communication technology infrastructure has significant positive impact on academic staff job performance in public universities in South-East, Nigeria. This finding is justified by the response patterns to questionnaire items 9–16, which reveal that lecturers consistently agreed that reliable internet connection enhances access to academic resources, updated computer systems improve lecture

preparation, ICT tools facilitate virtual teaching, e-resources help update course materials, online assessment tools support student evaluation, technical support accelerates problem resolution and ICT infrastructure boosts research output. These indicators show that ICT resources are now inseparable from modern teaching, research and administrative tasks. This finding aligns with Oladipo, Ajayi and Oyetunde (2021), who reported that ICT facilities such as multimedia classrooms, internet connectivity and e-libraries significantly improved teaching efficiency, communication and research productivity among academic staff in South-West, Nigeria. Their study supports the present one by demonstrating that ICT availability is a major determinant of lecturer productivity, although their study focused on utilization rather than provision a conceptual gap addressed by the present study. Similarly, Nwankwo, Okeke and Udeh (2022) found that ICT resources significantly influenced lecturers' academic performance in federal universities in South-East, Nigeria, particularly in access to publications and delivery of online lectures, making their findings highly relevant due to the shared geographical context and similar variable focus. However, while they focused on federal universities alone, the present study incorporates both federal and state institutions, thus expanding institutional coverage. The consistency between these studies and the present findings reinforces the justification that ICT infrastructure is indispensable for effective academic work, as it enhances productivity, reduces instructional delays, supports blended learning environments and improves research visibility. Within the boundaries of the study, it is found that provision of information and communication technology infrastructure has significant positive impact on academic staff job performance in public universities in South-East, Nigeria. Within the boundaries of the study, it is found that provision of information and communication technology infrastructure enhances academic staff job performance in public universities in South-East, Nigeria.

Conclusion

This study established that school plant management positively impact academic staff job performance in public universities in South-East, Nigeria. It was therefore concluded that school plant management in the areas of research facilities, information and communication technology infrastructure have significant positive impact on academic staff job performance in public universities in South-East, Nigeria.

Recommendations

Based on the findings of the study, the following recommendations were made:

1. University administrators should upgrade research facilities by procuring modern laboratory equipment and improving research support units through strategic budgeting. Better research environments will strengthen staff output, innovation and scholarly performance.
2. The University management should ensure that University ICT departments infrastructure is expand by upgrading internet bandwidth, digital tools and e-learning systems through sustained funding and regular maintenance. Enhanced ICT access will improve teaching, research efficiency and academic staff performance.

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